INCH-POUND

MIL-PRF-44259D
w/AMENDMENT 1
16 MARCH 2004
SUPERSEDING
MIL-L-44259D
9 August 2002

PERFORMANCE SPECIFICATION

LIGHT SET, PORTABLE, FLUORESCENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of portable fluorescent lights.

1.2 Classification. The light sets will be of the following types:

Type I - Set of four lights for Tent, Extendable, Modular, Personnel (TEMPER)
Type II - Set of two lights for Tent, Modular, Command Post, System (MCPS)

1.3 Part or identifying number (PIN). A document based PIN to identify the type of light set is included in section 6 (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS
MILITARY

MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: Defense Supply Center Philadelphia, ATTN: DSCP-ITAA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-4-4005	-	Strap, Light Support Assembly, Type I
5-4-6721	-	Strap, Light Support Assembly, Type II Light
5-4-6774	-	Label, Wiring Diagram
5-4-6785	-	Label, Date of MFG
5-4-6822	-	Label, Caution & ID

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted will be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, I Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

C82.11 - For Lamp Ballasts, High Frequency Fluorescent Lamp Ballasts - Supplements

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL943 - UL Standard for Safety for Ground-Fault Circuit-Interrupters

(Application for copies should be addressed to the Underwriters Laboratories, Inc. (UL), 333 Pfingsten Road, Northbrook, IL 60062-2096.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 First article. When specified (see 6.2d), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer provided the complete item complies with all provisions of this document. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

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3.2.1 Alternate components. Components shall be as specified herein. For components for which it is stated "or equal", if the contractor proposes to use an item considered to be equal to the component specified, prior to its use the contractor shall furnish a sample of the component, with supporting data to the contracting officer for subsequent evaluation by the responsible military agency. The supporting data required shall prove the functional equivalence and the design compatibility of the item proposed to be used. Prior to manufacture of the first article, the contractor shall submit for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with the requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of the first article or production manufacture, as applicable, and does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.3 Design and construction.

3.3.1 Light Set. Lights shall be capable of being interconnected with up to 11 other lights of the same design. Each light shall have handle(s) or end cap(s) on both ends that provide for the water resistant pass through of fixed external power cords. The light shall have a fixed power cord at one end of the light, which has a NEMA 5-15 plug. The light shall have a fixed power cord at the other end of the light, which has a NEMA 5-15 receptacle. A clearly marked on/off switch shall be provided on the end cap, or handle, that is on the same side of the light as the NEMA 5-15 plug. The power cords, plug, receptacle and end caps or handles shall be flat black in color. The combined length of the light fixture and input and output power cables shall be no less than 113 inches and no more than 120 inches. Each light shall weigh no more than 6 pounds and shall have a cube of less than .30 cubic feet.

3.3.1.1 Type I. The type I light set shall include four individual fluorescent light assemblies with filters (see 6.2e) when specified and four light support straps in accordance with drawing 5-4-4005.

3.3.1.2 Type II. The type II light set shall include two individual fluorescent light assemblies with filters and four light support straps in accordance with drawing 5-4-6721.

3.4 Electrical characteristics.

3.4.1 Safety. The requirements for elements of safety for items specified shall implement and conform to the safety standards issued by the Occupational Safety and Health Administration (OSHA) under the Secretary of Labor. Design and assembly of each light assembly shall comply with the applicable requirements of NFPA No. 70.

3.4.2 Input voltage. The light assembly shall operate on a voltage of 120 plus or minus 5 volts alternating current at 50 to 60 Hertz (Hz) ±5 Hz.

3.4.3 Input current. The input current drawn per light assembly shall be a maximum of 0.7 amp, when tested as specified in 4.5.5.

3.4.4 Electrical switch. The light assembly shall be designed so that power will pass through the light assembly to the next light assembly when connected in series regardless of the position of the ON/OFF switch.

3.4.5 Ballast. Lights shall operate through a ballast that is designed to provide thermal protection, by inherently limiting case temperatures, or by the use of a reset able fuse. The ballast temperature limiting is 158^0F when tested as specified in 4.5.6. The temperature rating shall be consistent with the capabilities of material and components in the ballast. The ballast shall be designed and assembled to conform to ANSI C82.11 unless requirements in this specification state otherwise. The ballast shall operate on either 50 or 60 Hz ±5 Hz, and shall function properly when tested as specified in 4.5.1 through 4.5.16.

3.4.6 Interconnection of lights. The lights shall be of a design to provide for interconnecting a string of up to 12 lights maximumwhile plugged into a Ground Fault Interrupt Circuit meeting requirements of UL 943. Feed through power connectors and wiring shall be rated for a 10 amp load. Each light shall operate independently on the through power circuit. Individual lights and lights connected in series shall be capable of operating without tripping the GFCI when tested as specified in 4.5.16.

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3.4.7 Dielectric strength. The light assembly shall be capable of withstanding 1,000 volts AC, 60 Hz between supply terminals (black and white) and with ground lead (green) grounded for a maximum period of 15 seconds when tested as specified in 4.5.7.

3.4.8 Electromagnetic capability. The complete light assembly shall meet the basic electromagnetic interference limits as specified under test method RE 102 for Army ground applications and CE 102 of MIL-STD-461 when tested as specified in 4.5.1.

3.4.9 Continuity-insulation resistance. Each light assembly shall have a direct current (DC) resistance of less than 1 ohm when tested for electrical continuity as specified in 4.5.8. In addition, insulation resistance shall be greater than 100 mega-ohms when tested as specified in 4.5.9.

3.4.10 Illumination. An individual light shall provide a minimum illumination of 140 foot candles when measured 18 inches from the surface of the assembly at the midpoint of the assembly using a cosine corrected photometer probe when tested as specified in 4.5.4.

3.4.10.1 Filter. The blue filter shall reduce the illumination of the light assembly to 15 – 25 foot candles when measured 35.00 ±1/4 inches from the surface of the assembly at the midpoint and at either end using a cosine corrected photometer probe when tested as specified in 4.5.4.1. The filter shall meet the chromaticity requirements of u'=.180, v'=.500, and r'=.055 on figure 1. No more than .5 percent of the total energy emitted shall be above 700 manometers and begin as close to 600 nanometers as possible. Ninety-five percent of the total energy should be confined within 400-600 nanometers.

CIE 1976 UCS DIAGRAM

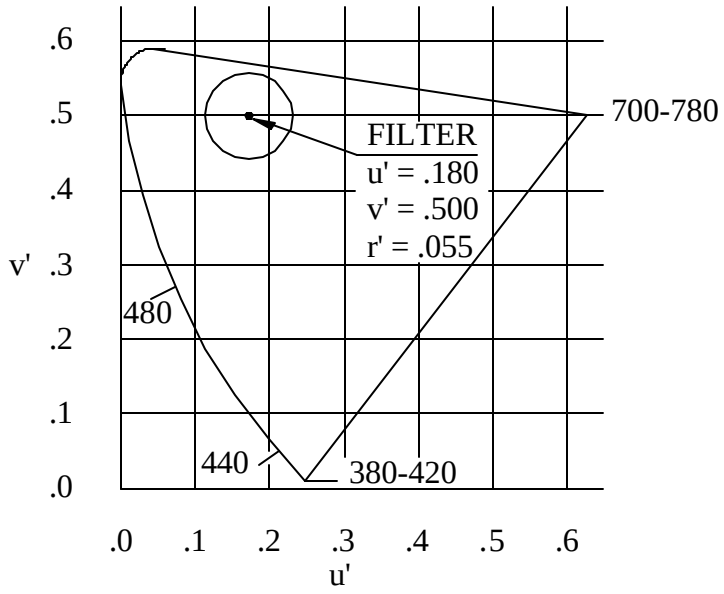

FIGURE 1. 1976 UNIFORM CHROMATICITY SCALE DIAGRAM SHOWING ALLOWABLE COLOR CIRCLE FOR FILTER

3.5 Performance.

3.5.1 High and low temperature storage. The sample light assemblies shall show no evidence of parts chipping, cracking, discoloration, extruding of impregnating or potting compounds, warping, deformation, or other defects detrimental to the intended function when tested for high temperature storage and cold temperature storage as specified in 4.5.10.

3.5.2 High temperature operation. Each of the light assemblies from the sample light sets shall show no evidence of parts chipping, cracking, warping, deformation, improper operation, failure to stay illuminated, or other defects detrimental to their intended function when tested for high temperature operation as specified in 4.5.11.

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3.5.3 Humidity. The sample light assemblies shall show no evidence of swelling of materials due to moisture absorption, degradation of electrical properties in insulating materials, electrical shorts, oxidation, galvanic corrosion, failure of lamps to light, or other defects which are detrimental to the intended function when tested for humidity as specified in 4.5.12.

3.5.4 Vibration. The sample light assemblies shall show no evidence of wire chafing, loose parts, touching or shorting of electrical parts, component fatigue, cracking of material, permanent deformation, failure of lamps to light, or other defects detrimental to the intended function when tested for vibration as specified in 4.5.13.

3.5.5 Noise levels. Each of the light assemblies from the sample light set shall be operated in series. The audible noise shall not exceed 50 decibels at the A scale sound performance level (SPL) at 1 foot from the light unit when tested as specified in 4.5.14.

3.5.6 Low temperature performance. Each of the light assemblies from the sample light set shall be capable of starting at 0^0F when tested as specified in 4.5.2 and 4.5.3. The light assembly shall start within 15 seconds, continue illumination, and properly operate for 1 hour after start up. Reduced light output is acceptable during this test.

3.6 Interchangeability. All light assemblies containing like part like numbers manufactured to the requirements of this document shall be physically and functionally interchangeable without selection or fitting when tested as specified in 4.5.15.

3.7 Labels. The labels shall conform to Drawings 5-4-6774, 5-4-6785, and 5-4-6822. The contents of the labels shall be clearly legible and shall be firmly affixed to the light set in the required locations with no areas of poor or no adhesion. The labels shall be free of tears and scratches.

3.8 Workmanship. Light assemblies shall be manufactured and processed in such a manner as to be uniform in quality and free from defects that will affect life, serviceability, or appearance. Metal components shall be free of burrs and sharp edges. Assemblies shall not contain loose, broken, inoperative, or otherwise damaged parts. Parts and assemblies shall be clean and free of oils, solder fluxes, and other extraneous matter; threaded parts shall show no evidence of looseness, cross threading, or stripping; wire insulation shall show no evidence of burns, abrading, or pinch marks; and solder connections shall have a smooth bright appearance with metallic luster and shall not have a chalky, gritty, or irregular surface nor exhibit any protrusions, pits, or voids. Finish shall be smooth and uniform without sags, pinholes, blisters, scratches, wrinkles, foreign matter imbedded in finish, or tackiness. The finished light assembly shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 3.1 and 6.2d), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and shall be tested for the characteristics specified in table I and 4.4.5.

TABLE I. First article tests

Characteristic	Requirement paragraph	Test method
Electromagnetic compatibility	3.4.8	4.5.1
Illumination	3.4.10	4.5.4
Light reduction	3.4.10.1	4.5.4.1
Cold starting at 60 Hz	3.5.6	4.5.2
Cold starting at 50 Hz	3.5.6	4.5.3
Thermal protector	3.4.5	4.5.6
Interconnection of lights	3.4.6	4.5.16
Dielectric strength	3.4.7	4.5.7
Insulation resistance	3.4.9	4.5.9
High and low temperature storage	3.5.1	4.5.10
High temperature operation	3.5.2	4.5.11
Resistance to humidity	3.5.3	4.5.12
Resistance to vibration	3.5.4	4.5.13
Noise levels	3.5.5	4.5.14
Interchangeability	3.6	4.5.15

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-1916.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details, which cannot be examined in the finished product, are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance. Inspection shall be made of the following fabrication operations to establish conformance with specified requirements:

a. That color coded wire is used throughout the wiring of the light assembly and is in agreement with the wiring schematic and that the wire is of the specified wire size.

b. That all materials are properly coated or protected from corrosion effects.

c. That all electrical wires are properly routed and insulated to prevent abrasion from sharp edges.

d. That all mounting hardware is properly secured.

e. That all soldering is in accordance with specified requirements and soldered connections have complete fusion, no excess solder, no porosity or inclusions, and all flux has been removed.

f. That any crimp-type electrical fasteners are secure on the wires and that no wire strands are broken or not secured within the fastener.

g. That wiring schematic is displayed in ballast chamber.

Whenever nonconformance is noted, correction shall be made to the item affected and the process. Items, which cannot be corrected, shall be removed from production.

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4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II, The lot size shall be expressed in units of complete light sets. The sample unit shall be one complete light set. The verification levels shall be IV for major defects and II for minor defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not adherent, e.g., blistered, peeled, or flaky		201
	Coating not continuous or uniform i.e., not coated or area of no film		202
	Scratch, dig, or abrasion exposing bare metal or primer		203
Color	Not color specified	101	
Electrical cable	Not proper size or type	102	
	Cut or abrasion	103	
Electrical connector	Missing or not correct part number	104	
Light straps	Not as specified	105	

TABLE II. End item visual defects (cont.)

Examine	Defect	Classification Major	Minor
Design and Workmanship	Sharp edge or burr	106	
	Puncture, fracture, or malformed	107	
	Design detail deviating from document	108	
	Evidence of rust or deposit of any foreign matter		204
Light switch	Missing, in-operable, or not correct switch	109	
Light switch weather boot or covering	Missing, ripped, torn or cracked		205
Markings or labels	Omitted, incorrect, illegible, misplaced		206

4.4.4 End item dimensional examination. The light set shall be examined for dimensional defects listed in table III. The lot size shall be expressed in units of complete light sets. The sample unit shall be one complete light set. The verification levels shall be IV for major defects and II for minor defects.

TABLE III. End item dimensional defects

Examine	Defect	Classification Major	Minor
Overall dimensions (individual light applicable minus tolerance assemblies)	Smaller than specified less	101	
	Larger than specified dimensions and applicable plus tolerances	102	
Overall dimensions	Dimensions not as specified	103	

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4.4.5 End item testing. The complete light set shall be tested for the characteristics listed in table IV. The lot size shall be expressed in terms of complete light sets. The sample unit shall be one complete light set. The verification level shall be IV.

TABLE IV. End item tests

Characteristic	Requirement paragraph	Test method
Current draw	3.4.3	4.5.5
Electrical continuity	3.4.9	4.5.8

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The verification level shall be IV.

Examine	Defect
Marking (interior and exterior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application
Materials	Any component missing, damaged, or not as specified

Examine	Defect
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling
Content	Number of light sets per shipping lot is more or less than required

4.5 Methods of inspection.

4.5.1 Electromagnetic compatibility test. Each of the light assemblies from the sample light set shall be tested in accordance with test methods CE 102 and RE 102 of MIL-STD- 461. Failure of any light assembly to meet the limits specified in MIL-STD-461 (see 3.4.8) shall constitute a test failure.

4.5.2 Cold starting test at 60 Hz. Each of the light assemblies from the sample light set shall be cold soaked for 4 hours at a temperature of 0^0F. While maintaining a temperature of 0^0F, an electric power of 120 $\pm$ 5 volts, 60 Hz shall be applied to each light assembly and the light operated for a minimum of 1 hour. Failure of any light assembly to properly operate or conform to the requirements of 3.5.6 shall constitute a test failure.

4.5.3 Cold starting test at 50 Hz. Each of the light assemblies from the sample light set shall be cold soaked for 4 hours at a temperature of 0^0F. While maintaining a temperature of 0^0F, an electric power of 120 $\pm$ 5 volts, 50 Hz shall be applied to each light assembly and the light operated for a minimum of 1 hour. Failure of any light assembly to properly operate or conform to the requirements of 3.5.6 shall constitute a test failure.

4.5.4 Illumination test. Each light assembly from the sample set shall be tested for illumination when operating at 120 ± 5 volts, 60 Hz, and a maximum current of 0.7 amperes in an atmosphere of 77 ± 5^0F. Failure to provide illumination as specified in 3.4.10 shall constitute a test failure.

4.5.4.l Filter test.. Each filter from the sample light set shall be tested for light reduction when the light assembly is operating under conditions specified in 4.5.4. Failure to provide the conditions specified in 3.4.10.1 shall constitute a failure.

4.5.5 Current draw test. Each light assembly from the sample light set shall be tested for current draw when operating at 120 ± 5 volts, 60 Hz. Any light assembly exceeding 0.7 amp, as specified in 3.4.3, shall constitute a test failure. Repeat this test at 120 ± 5 volts and 50 Hz.

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4.5.6 Thermal protector test. One ballast assembly (or a represented standard ballast circuit) shall be tested for thermal protection. Place ballast, which operates one external lamp at 120 V AC, 60 Hz inside an oven or similar chamber. Place a thermocouple on the ballast case. Soak the operating ballast for a minimum of 2 hours at 140^0F and record stabilized case temperature and input current. Increase ambient temperature at 10 degree increments allowing case to stabilize at each increment until 160^0F has been reached. Record case and ambient temperatures and input current at each increment. Return chamber to 140^0F and observe normal operation. Record input current. Failure to conform to the requirements of 3.4.5 shall constitute a test failure.

4.5.7 Dielectric protector test. One light assembly from the sample light set shall be tested for a dielectric load for the period as specified in 3.4.7. Any inability to accept the specified voltage for this minimum time period without disruptive discharge or any current leakage shall constitute a test failure.

4.5.8 Electrical continuity for light assemblies. Each conductor in the electrical light assembly shall be tested in accordance with Method 303 of MIL-STD-202. The test current shall not exceed 15 milliamperes and the test resistance shall not exceed one ohm. Measurement shall be made between each connector pin, which is mutually common to each other electrically. The continuity for each pin shall be checked for compliance with the continuity requirement of 3.4.9 and failure to conform to continuity requirement shall constitute a test failure.

4.5.9 Insulation resistance for light set assemblies. With the light switch "OFF", the light set assembly shall be tested in accordance with Method 302 of MIL-STD-202. The test potential shall be 500 volts, test condition B. Insulation resistance measurement shall be made between the mutually insulated points and between insulated points and shell or ground. Each resistance measurement shall be checked for compliance with the insulation resistance requirement of 3.4.9. Failure to conform to the continuity requirement shall constitute a test failure.

4.5.10 High and low temperature storage test. The sample light assemblies in their packaged form shall be tested for high temperature storage in accordance with Method 501, procedure I of MIL-STD-810. The test temperature and cycle shall be as specified under hot induced conditions with a minimum of seven complete cycles. The sample light set shall then be tested for low temperature storage in accordance with Method 502, procedure I of MIL-STD-810. The test temperature shall be severe cold (C3) (-60^0F) with the duration of exposure at 8 hours. After each storage test the sample light assemblies shall be removed from the test chamber, allowed to stabilize at room temperature, then operated. Failure to conform to the requirements of 3.5.1 shall constitute a test failure.

4.5.11 High temperature operation test. Each of the sample light assemblies shall be tested for high temperature operation in accordance with method 501, procedure II of MIL-STD 810. The test temperatures and cycle shall be as specified under hot, ambient air conditions. The light assemblies shall be hooked up in series and shall be operated through three complete temperature cycles. Failure to conform to the requirements of 3.5.2 shall constitute a test failure.

4.5.12 Humidity test. The sample light assemblies shall be tested for resistance to humidity with the light assemblies set up in series and exposed to chamber conditions. The test shall be conducted in accordance with Method 507, procedure I, cycle 2 (quick look) of MIL-STD-810. Operational checks are required during testing as specified in MIL-STD-810. During operational checks, the light assemblies shall be operated for 1 hour under chamber conditions. Failure to conform to the requirements of 3.5.3 shall constitute a test failure.

4.5.13 Vibration test. The sample light assemblies shall be tested for resistance to vibrations encountered during transport in the field. The light assemblies shall not be packaged. The test shall be conducted in accordance with Method 514, category 3 (loose cargo transport), with test duration of 3 hours as specified in MIL-STD-810. Upon completion of vibration test remove the light assemblies, hook up the lights in series and operate for 10 minutes. Failure to conform to the requirements of 3.5.4 shall constitute a test failure.

4.5.14 Noise test. Each light assembly from the sample light set shall be tested for noise by operating the unit at the specified voltage and input current and at 50 and 60 Hz. Using a suitably calibrated instrument, the noise at 1 foot from the light assembly, in the area of the ballast, shall be recorded in at least three perpendicular planes around the circumference of the light assembly. Failure to conform to the requirements of 3.5.5 shall constitute a test failure.

4.5.15 Interchangeability test. The light assemblies shall be tested for interchangeability by selecting two light assemblies from the light set and interchanging like part numbered components that would be considered repair or replaceable parts. After interchanging the parts, the light assemblies shall be operated at the rated voltage, amperage, and either 60 Hz or 50 Hz. Any inability to interchange, (see 3.6) or for the light assemblies to properly operate, shall constitute a test failure.

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4.5.16 Interconnection test. The light assemblies shall be tested for inter-connectivity with eleven other lights and compatibility with GFCI. Start the test with a single light assembly plugged into a 20 amp UL 943 rated GFCI. The light assembly shall be checked to insure that it operates properly in the on position without tripping the GFCI. A second light will be added in series to verify that both lights operate properly without tripping the GFCI. The first light in series will be turned off while the second light is left on to verify that each light operates independently on the through power circuit. Failure of the next light in series to remain illuminated would constitute a failure. This process will be repeated until all twelve lights are interconnected in series without tripping the GFCI. As each light is added in series the previous light in series will be turned off to verify the through power circuit. Failure to conform to the requirements of 3.4.6 shall constitute a test failure.

4.6 Verification Alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost affective sampling procedures to verify performance. See the contract for approved alternatives that replace verifications required by this specification.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2f). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

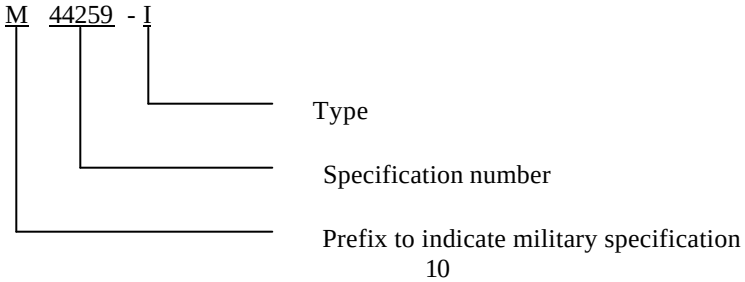
6.1 Intended use. The light sets are designed to provide high intensity lightweight lighting, of flexible usage, rugged construction and have easily replaceable lamps. The type I light set consists of four light assemblies and is to be used with the Tent, Extendable, Modular, Personnel (TEMPER). The type II light set consists of two light assemblies and is to be used with the Tent, Modular, Command, Post, System (MCPS).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of light set required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3)
d. When a first article is specified (see 3.1, 4.3 and 6.3).
e. When lamp filters are required for the type I light set (see 3.3.1.1)
f. Packaging requirements (see 5.1).

6.3 First article. When a first article is required, it should be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be a pre-production sample of three Type I sets or six Type II sets. These quantities are needed to perform 4.5.7 Interconnection test. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Part or identification number (PIN). The PIN's to be used for light sets acquired by this specification are created as follows:



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6.5 Subject term (key word) listing.

Lamps
TEMPER
MCPS

6.6 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - GL
Air Force - 99

Preparing activity:
DLA - IS

Review activities:
Air Force - 84

(Project 6230-1088)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

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